

Midterm Study Guide

POLS 602

These questions cover the conceptual ideas from the first half of the course. Try to answer each one in your own words. Many questions ask you to explain why, and the exam will too.

Causal Inference and Experiments

1. Define the treatment variable and the outcome variable. Give examples.
2. What is the fundamental problem of causal inference? How does this relate to counterfactuals?
3. What is the average causal effect? Why is it easier to estimate than an individual effect, and what assumption(s) does it require?
4. What does random assignment accomplish, and why does the sample size matter?
5. Write the difference-in-means estimator. Why does it have a hat? In an experiment the control group voted at a rate of 0.30 and the treatment group at 0.38. What is the estimate, and how would you interpret it?
6. What distinguishes an observational study from a randomized experiment? What must you worry about in an observational study that you do not in a randomized one?

Dot Products, Norms, and Moments

7. What is the dot product of two vectors? What must be true of the vectors, and what kind of object does it return?
8. When is the dot product of two vectors zero? What does this mean geometrically? What is the relationship between covariance and the dot product?
9. What is a norm? What is the L2 norm?
10. What is a moment? Which moments correspond to the mean, variance, and skewness? What is a central moment?
11. What is the difference between a PDF and a PMF? What do density and mass represent in each?
12. Why can the mean be a poor description of a typical value? Give two examples of how the mean can mislead. Why is the median called robust?
13. Define variance and standard deviation. Why do we usually report the standard deviation? Why does the sample variance divide by $n - 1$ instead of n ?
14. What is a z-score, and why is it useful for communicating how far a value is from the mean? An observation is 3 standard deviations above the mean. What does that tell you?
15. What does skewness measure? Describe right skew, and give an example of a variable that is likely to be right skewed. How do the mean and median compare in that case?

16. Define covariance and the Pearson correlation. What does the sign tell you? What does the magnitude tell you? Why is the correlation easier to interpret than the covariance?

17. Explain the notational difference between μ , $\bar{x}$, $\hat{\mu}$, and $E[Y]$. What does $E[Y | X]$ mean?

Prediction and Simple Regression

18. What does it mean to say a model is wrong but useful? Explain why criticizing a model's assumptions is not enough to reject it.

19. Distinguish the true regression model $Y_i = \alpha + \beta X_i + \varepsilon_i$ from the estimated model $\hat{Y}_i = \hat{\alpha} + \hat{\beta} X_i$. Which quantities are observed? What is the difference between ε_i and $\hat{\varepsilon}_i$?

20. What does a point on the regression line represent? Interpret $\hat{\alpha}$ and $\hat{\beta}$.

21. Consider a bivariate regression where X is binary. What does $\hat{\alpha}$ represent? What does $\hat{\beta}$ represent? What estimator is $\hat{\beta}$ the same as, and why does that matter?

22. What is the difference between an estimand, an estimator, and an estimate? :::: {content-visible when meta="solutions"} :::: {solution} The estimand is the quantity we want to learn (the average treatment effect). The estimator is the rule for computing it from data (the difference in means). The estimate is the number the rule produces for a particular sample (0.08). :::: ::::

23. Write the design matrix $\mathbf{X}$ for a simple regression with n observations. Why does it have a column of ones? What are its dimensions, and what are the dimensions of $\hat{\beta}$ and $\hat{\mathbf{Y}}$?

24. Define the sum of squared errors in scalar and matrix form. What does it measure? Why do we square the residuals?

Ordinary Least Squares

25. What is OLS? Explain how it works conceptually.

26. Why does setting the derivative of SSE to zero find the best line? Why do we need partial derivatives? Why is the flat spot a minimum and not a maximum?

27. The derivative of SSE with respect to the intercept, set to zero, says something about the residuals. What? Explain what this implies about where the line passes.

28. The derivative of SSE with respect to the slope, set to zero, says something about the residuals and X . What? Why is this the right condition for the best slope?

29. State the OLS slope as covariance over variance. Why do we divide by the variance of X ?

30. Two datasets have the same covariance between X and Y , but X is more spread out in the second. Which has the steeper slope, and why?

31. How does the slope relate to correlation?

32. You have $s_{XY} = 6$, $s_X^2 = 3$, $\bar{X} = 4$, and $\bar{Y} = 10$. Compute $\hat{\beta}$ and $\hat{\alpha}$. Predict Y at $X = 5$.

33. State the normal equations and the OLS solution. Which step of the derivation requires the inverse to exist?

34. Describe OLS geometrically. What is the column space of $\mathbf{X}$? What is $\hat{\mathbf{Y}}$ relative to $\mathbf{Y}$? Why is the residual perpendicular to the column space?

35. List the three mechanical properties of the OLS line that hold regardless of whether any assumptions are true. Why can you not use them to check whether the model is correct?

Multiple Regression

36. Why add multiple predictors to a regression model? Provide an example to illustrate why this might be important.

37. Define bias. Is bias a property of a single estimate or of an estimator? Why does collecting more data not fix omitted variable bias?

38. What is a confounder? Does an omitted variable that affects Y but is unrelated to X bias the coefficient on X ? What about one that is related to X but does not affect Y ?

39. What does it mean to hold a variable constant in multiple regression? Give both a conceptual meaning and the mathematical one.

40. Write the design matrix for a regression with k predictors. What are its dimensions, and what changes in the matrix form $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ compared to the simple regression?

41. What is the dummy variable trap? Why does it violate a Gauss-Markov assumption?

42. Write a model with an interaction between X_1 and X_2 . What does β_3 mean? What is the effect of X_1 on Y in this model?

43. Why is a squared term a kind of interaction? What does it let the model do that a single linear term cannot? Is a model with X^2 still a linear model?

Gauss-Markov and the Properties of OLS

44. What does BLUE stand for? Explain what each word means.

45. BLUE is a claim about the estimator, not about a single estimate. What is the difference? If you run OLS on your data, is your estimate guaranteed to be unbiased?

46. Can an estimator be biased but have low variance? Unbiased but have high variance? Why are bias and variance separate problems?

47. What does it mean for a model to be linear in its parameters? Which of these are linear in the parameters, and why?

(a) $Y_i = \beta_0 + \beta_1 X_i^2 + \varepsilon_i$

(b) $Y_i = \beta_0 + \beta_1 \log(X_i) + \varepsilon_i$

(c) $Y_i = \beta_0 + X_i^{\beta_1} + \varepsilon_i$

(d) $Y_i = \frac{\beta_0 X_i}{\beta_1 + X_i} + \varepsilon_i$

48. State the four Gauss-Markov assumptions. For each, explain its significance.

49. Is normality of the errors a Gauss-Markov assumption? What is it used for?

50. Why can you not test strict exogeneity by checking whether the residuals are correlated with X ?

51. Provide an example of how exogeneity can fail. What does it mean to say a predictor is endogenous?

52. In the omitted variable bias formula, what is the interpretation of $(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$? Conceptually, how can we think about the way an omitted variable biases our estimate?

53. A study finds that a news consumption coefficient on political knowledge is positive and significant, and you worry political interest was omitted. Which direction is the bias likely to push the estimate, and why? How does your answer change if interest and news consumption were uncorrelated?
54. What does full column rank mean? Give a concrete example of a matrix that violates it, and explain why the coefficients cannot be identified.
55. What is homoscedasticity? What does a plot of heteroscedastic data look like? :::: {content-visible when-meta="solutions"} :::: {solution} The errors have the same variance at every value of $\mathbf{X}$, $\text{Var}(\varepsilon_i | \mathbf{X}) = \sigma^2$. Heteroscedasticity is nonconstant variance, for example scatter that fans out as X grows. Errors are spherical, $\text{Var}(\varepsilon | \mathbf{X}) = \sigma^2 \mathbf{I}$, when they also are uncorrelated with each other (no autocorrelation). ::::
56. If errors are heteroscedastic, is OLS still BLUE?
57. Which Gauss-Markov assumption is the hardest to defend in observational research, and why? Which can your software check for you? Which has a standard fix?
58. For each situation, name the Gauss-Markov assumption that is most likely violated.
- (a) The standard errors in a regression table are too small, but the coefficients look right.
 - (b) The coefficient on X is off, and collecting ten times more data does not help.
 - (c) The software drops a variable from the model.

Assumptions for Inference

59. The Gauss-Markov assumptions describe the estimator across repeated samples. What three additional assumptions do we need to learn about a population from a single sample?
60. What is validity? Give an example of an invalid measure. Why can a regression give a precise, unbiased estimate of the wrong thing?
61. What is attenuation bias? What happens to the slope when the predictor is measured with random noise? What about random noise in the outcome? Which Gauss-Markov assumption does noise in the predictor violate?
62. What is representativeness?
63. Compare selecting a sample on X with selecting it on Y . Which biases the slope, and why?
64. State the normality assumption and the resulting sampling distribution of $\hat{\beta}$. What does normality buy us? Why does it matter less in large samples?
65. Suppose you read a study that regresses a measure of legislative effectiveness on a measure of legislator ideology using data from a single Congress, and finds a positive coefficient. Identify one way each of the following could undermine the conclusion, and say whether the problem affects bias, variance, or inference: (a) omitted variables, (b) an invalid effectiveness measure, (c) a sample that includes only legislators who ran for reelection, (d) heteroscedastic errors.